

# Scoreboard

A scoreboard checks that what you predicted matches what the DUT actually did

# Predict, observe, compare

- The predict path builds what should happen
- The observe path is the monitor analysis port, real bus activity becomes actual
- Compare is the heartbeat: pop one from each queue, check equality, repeat
- A pass means the values matched; a fail means expect and actual differed
- Orphans are leftovers, expect with no actual means you predicted something never observed
- Both are usually bugs at end of test

# Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

## Scoreboard expect/actual

Sketch the compare pipeline: **predict** → **expect queue**, **observe** → **actual**, then **match** / **mismatch** / **orphan**. Starter: one expected and one actual both 0xA5 — PASS.

Starter example: expect and actual both hold 0xA5 — Compare should PASS.

Load starter example

Challenges 0 / 22 cleared

Quiz: role: A scoreboard's main job is to...

☐ compare predicted (expect) vs observed (actual) transactions

☐ synthesize the DUT netlist

☐ replace the sequencer entirely

☐ set the simulator timescale

Show hint

Check

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Idle

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Core ideas

expect

Predicted transaction from a model or stimulus mirror.

actual

Observed transaction from the monitor.

compare

Pop fronts; match or report mismatch.

orphans

Leftover expect or actual at end of test is a bug.

Lab

SCENARIO

TXN DATA

Scoreboard lab starter

learn\_uvm2017 — uvm scoreboard

# Real UVM literacy

```
❯ uvl - module13-uvm-scoreboard/examples/scoreboard-sketch

# Track A - Real Unix
# real shell session (Track A - UVM scoreboard expect/actual)

$ pwd
/mnt/d/proj/designs/learning/courses/Learn_uvm2017/module13-uvm-scoreboard

$ ls examples/scoreboard-sketch/
compare.txt

$ cat examples/scoreboard-sketch/compare.txt
UVM scoreboard sketch (Literacy)

PipeLine:
predict/reference  + expect queue
monitor analysis  + actual queue
compare           + pop fronts, check match

Conceptual compare loop:
while (expect_q.size() && actual_q.size()) begin
  exp = expect_q.pop_front();
  act = actual_q.pop_front();
  if (!exp.compare(act))
    `uvm_error("SB", "mismatch")
end
// end of test: leftover queues = orphans

Typical UVM wiring:
`uvm_analysis_imp_declf_expected
`uvm_analysis_imp_declf_actual

class my_scoreboard extends uvm_scoreboard;
  uvm_analysis_imp_expected #(my_txn, my_scoreboard) expected_imp;
  uvm_analysis_imp_actual  #(my_txn, my_scoreboard) actual_imp;
  my_txn expect_q[$], actual_q[$];

  function void write_expected(my_txn t);
    expect_q.push_back(t);
  endfunction

  function void write_actual(my_txn t);
    actual_q.push_back(t);
    try_compare(); // or compare in write
  endfunction
endclass

Connect phase:
ref_model.analysis_port.connect(sb.expected_imp);
monitor.analysis_port.connect(sb.actual_imp);

Verdicts:
PASS    - fronts match, both popped
FAIL    - fronts differ
ORPHAN  - one queue empty, other not (at compare or end of test)

Common mistakes:
compare without data on both sides
ignore leftover queues at test end
in-order compare on out-of-order protocol (needs different SB)

$ grep -n "scoreboard\\|write_expected\\|write_actual\\|expected_queue\\|actual_queue"
../Learn_uvm2017_sv_verilator/modules/tests/uvm_tests/test_utilities_uvm.sv | head -12
121: * Scoreboard using comparator
123: class UtilitiesScoreboard extends uvm_scoreboard;
124:   UtilitiesTransaction expected_queue[$];
125:   UtilitiesTransaction actual_queue[$];
129:   uvm_analysis_imp_expected #(UtilitiesTransaction, UtilitiesScoreboard) expected_imp;
130:   uvm_analysis_imp_actual  #(UtilitiesTransaction, UtilitiesScoreboard) actual_imp;
132:   `uvm_component_utils(UtilitiesScoreboard)
134:   function new(string name = "UtilitiesScoreboard", uvm_component parent = null);
142:   function void write_expected(UtilitiesTransaction txn);
143:     expected_queue.push_back(txn);
147:   function void write_actual(UtilitiesTransaction txn);
148:     actual_queue.push_back(txn);
```

Real shell — scoreboard sketch

# Pitfalls to watch

- Do not compare without both queues having data
- Do not ignore leftover queues at test end, orphans mean missing stimulus or missing checks
- Do not assume in-order compare works for every protocol
- And remember

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run starter compare pass then demo mismatch and name the verdict
- On real UVM, sketch predict and observe paths into one scoreboard
- When you are ready, take the short quiz, then continue to RAL map in the next module